

FELsim Beamline Optimisation: Technical Compendium

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11 March 2026

Abstract

We present a comprehensive summary of the FELsim transport line optimisation programme for the UH MkV free electron laser. The programme encompasses 28 completed studies across five categories: parameter sensitivity (S1–S9), cross-code validation (C1–C8), optimizer improvements (O1–O5), physics model upgrades (P1–P3), and infrastructure (I1–I8). We validated the 11-stage Nelder-Mead optimisation against COSY INFINITY (differential algebra) and RF-Track (particle tracking), achieving cross-code MSE agreement to 10^{-6} – 10^{-9} . A two-phase NM→CMA-ES polishing strategy improves Stage 11 precision by up to $10^{10}\times$. Chromatic physics (per-particle momentum-dependent matrices for quads, dipoles, and edge kicks) and aperture loss tracking were integrated into the production optimizer, raising simulated transmission from 64.5% to 98.1%. The feasibility boundary is dominated by emittance; energy spread and chirp have negligible impact on transverse matching at the baseline $\varepsilon_n = 8 \pi \cdot \text{mm} \cdot \text{mrad}$. The transport line cannot compress bunches below $\sim 0.45 \text{ ps}$ (limited by $R_{56} \times \sigma_\delta$); compression must occur upstream.

Contents

1	Introduction	3
2	Simulation Framework	3
2.1	FELsim (First-Order Transfer Matrices)	3
2.2	COSY INFINITY (Differential Algebra)	3
2.3	RF-Track (Particle Tracking)	4
2.4	Multi-Code Orchestrator (I7)	4
3	Transverse Beam Matching	4
3.1	Baseline Optimisation (S1, S3)	4
3.2	Cross-Code Validation (W4, C1, R2)	4
3.3	Algorithm Comparison	5
4	Parameter Sensitivity	5
4.1	1D Scans (S4)	5
4.2	2D Scans (S5)	6
4.3	Bunch Length Independence (S6, S9)	6
4.4	Verification (S7)	6

5	Longitudinal Dynamics	6
5.1	Transfer Map Analysis (W9)	6
5.2	Bunch Compression Feasibility (W12)	6
5.3	σ_z Blowup Investigation (I6)	6
6	Chromatic Optimisation	7
7	Beam Losses and Throughput	7
7.1	Loss Distribution (P1)	7
7.2	Throughput Optimisation (W11)	7
8	FR3+MGE Optimisation (C3)	7
9	Infrastructure	8
10	Summary of Findings	8
11	Study Index	8

1 Introduction

The UH Mark V FEL transport line consists of 118 beamline elements (23 quadrupoles, 4 dipole chicane sections, drifts, correctors, and diagnostics) that transport a 40 MeV electron beam from the linac exit to the undulator entrance. The beam must be matched to the undulator Twiss targets specified by Weinberg, Fisher & Li [1]:

$$\beta_x = 1.400 \text{ m}, \quad \alpha_x = 0.470, \quad \beta_y = 0.242 \text{ m}, \quad \alpha_y = 0. \quad (1)$$

FELsim decomposes this 23-variable matching problem into an 11-stage sequential optimisation, where each stage controls a physically motivated group of quadrupoles (singlets, doublets, triplets, chromaticity sections). The objective function is the weighted mean squared error (MSE) between computed and target Twiss parameters. Results are reported as the root mean square (RMS) $= \sqrt{\text{MSE}}$, which has the same units as the Twiss residuals.

This compendium summarises all completed studies, organised by topic rather than chronology, and provides cross-references to the detailed scripts, reports, and result files.

2 Simulation Framework

2.1 FELsim (First-Order Transfer Matrices)

FELsim’s native transport model uses 6×6 linear transfer matrices (`beamline.py`). Each element type has an analytical matrix: drifts, thick quadrupoles with cos/cosh focusing, sector-bend dipole bodies, and dipole wedges with Enge-function fringe corrections. The model evaluates in ~ 1 ms per beamline, enabling $\mathcal{O}(10^4)$ optimizer evaluations per study.

Chromatic extensions (P1–P3). All elements support an opt-in `chromatic=True` mode that computes per-particle momentum-dependent matrices:

- **Quadrupoles (C8):** $k_{\text{eff}} = k_0 \times \beta\gamma_0/\beta\gamma(\delta)$, matching RF-Track’s intrinsic momentum scaling to 0.00% across all Twiss parameters.
- **Dipole body (P3):** Per-particle $\rho_p = \rho_0 \times \beta\gamma_p/\beta\gamma_0$, $\theta_p = L/\rho_p$.
- **Dipole wedge (P2):** Momentum-dependent R , ϕ , and edge kicks ($h_p = h_0 \times P_0/P$).

All chromatic matrices are symplectic ($\det M = 1.0$ to machine precision) and vectorised over particles without Python loops.

Aperture loss tracking (P1). Physical apertures are defined per element type: QPF/QPD ± 13.5 mm (bore), DPH/DPW $\pm \text{pole_gap}/2$. The optimizer integration adds a smooth $W \times (1 - T)^2$ penalty to the MSE objective when aperture tracking is enabled.

2.2 COSY INFINITY (Differential Algebra)

COSY INFINITY computes transfer maps to arbitrary order using differential algebra (DA). The Python adapter (`cosyAdapter.py`) generates FOX code, executes the COSY binary, and parses output. Quadrupoles are thick MQ elements; dipoles are thick DIL with integrated edge angles and optional Enge fringe fields (FC/FD) or measured fieldmaps (MGE). The built-in FIT optimizer exploits DA derivatives for gradient-based matching.

2.3 RF-Track (Particle Tracking)

RF-Track (v2.5.5, CERN) performs element-by-element particle tracking with support for space charge. The adapter (`rftrackAdapter.py`) includes an analytical sector-bend workaround for the SBend P/δ bug (C5), thin-lens quadrupole edge kicks for DPW elements, and prefix caching for fast Stage 11 re-tracking.

2.4 Multi-Code Orchestrator (I7)

The `MultiCodeSimulator` chains multiple adapters on contiguous beamline sections, using FELsim coordinates as the interchange format. Validated configurations include FELsim→RF-Track, FELsim→COSY, COSY→RF-Track, and three-way FELsim→COSY→RF-Track hybrids.

3 Transverse Beam Matching

3.1 Baseline Optimisation (S1, S3)

The 11-stage Nelder-Mead (NM) optimiser achieves $MSE < 10^{-5}$ at the baseline operating point ($\varepsilon_n = 8 \pi \cdot \text{mm} \cdot \text{mrad}$, $\sigma_E = 0.5\%$, $h = 5 \times 10^9 \text{ s}^{-1}$, 0.5 ps bunch). The 2 ps (S1) and 0.5 ps (S3) optimisations produce identical quadrupole currents, confirming transverse-longitudinal decoupling at first order (S9).

3.2 Cross-Code Validation (W4, C1, R2)

Table 1 summarises the cross-code comparison at the baseline operating point.

Table 1: Cross-code optimisation at $\varepsilon_n = 8 \pi \cdot \text{mm} \cdot \text{mrad}$. Targets: $\beta_x = 1.400 \text{ m}$, $\alpha_x = 0.470$, $\beta_y = 0.242 \text{ m}$, $\alpha_y = 0$.

Code	Model	MSE
FELsim NM	Transfer matrix	1.2×10^{-6}
COSY FR 0	DA map (no fringe)	2.3×10^{-7}
COSY FR 1	DA map (1st fringe)	3.7×10^{-9}
RF-Track opt	Particle tracking	7.0×10^{-3}

FELsim and COSY achieve MSEs of 10^{-6} – 10^{-9} , matching all four Twiss targets to within 10^{-3} . COSY’s gradient-based DA optimizer converges $\sim 10^3 \times$ more precisely. RF-Track’s higher MSE (7.0×10^{-3}) reflects a systematic β_y deficit from the thin-lens edge-kick model (missing triangle-rule fringe correction ϕ), not optimizer limitations.

Bug fixes (R2). Achieving cross-code agreement required identifying and correcting 9 implementation errors: FELsim DPW edge-kick sign inversion, five RF-Track adapter issues (SBend P/δ confusion, DPW modelling, edge-kick sign, R-matrix units, coord5 conversion), and three COSY adapter issues (initial β_0 , FIT objective reset, warm-start for fringe fields). See the R2 report for details.

3.3 Algorithm Comparison

Nelder-Mead (NM). The baseline optimizer. Multi-start with 5 random Stage 11 initial conditions resolves local minimum traps at extreme emittances ($\varepsilon_n = 14\text{--}16$).

Glyfada (W6). The glyfada evolutionary optimizer (CMA-ES, NSGA-II) via the distributed DH evaluator protocol. At 600 evaluations per run, glyfada fails at all three test emittances ($\varepsilon_n = 5, 8, 14$); NM outperforms by 3–6 orders of magnitude in MSE. Root cause: uniform random initialisation over wide bounds wastes evaluations in infeasible regions.

CMA-ES polishing (W7, O5). pycma CMA-ES warm-started from the NM solution dramatically outperforms NM alone:

Table 2: Two-phase NM→CMA-ES vs NM-only for Stage 11 matching.

ε_n	NM MSE	CMA-ES MSE	Improvement
5	8.2×10^{-6}	4.5×10^{-16}	$\sim 10^{10} \times$
8	6.1×10^{-6}	1.2×10^{-15}	$\sim 10^9 \times$
14	8.9×10^{-3}	6.3×10^{-3}	$1.4 \times$

The two-phase strategy was integrated into the production optimizer (O5): after NM converges, CMA-ES refines from the NM solution with $\sigma = 0.1$, popsize= 20, maxfevals = 3000. The composite 5-goal objective (4 Twiss + dispersion) has a dispersion floor of $\sim 2.5 \times 10^{-6}$ that Stage 11 quads cannot reduce.

Stage 11 unimodality (S8). A multi-start robustness study (5 emittance points $\times$ 10 random starts) found that *all* random starts converge to identical solutions at machine precision. The Stage 11 landscape has a single global basin; multi-start cannot improve it. The seed-dependence observed at extreme emittances (S7) originates entirely from stages 1–10 (random beam generation affecting upstream matching).

Warm-starting (O1). Sequential warm-starting from neighbouring scan points is counterproductive: the $\varepsilon_n = 1$ basin (Marginal, RMS ≈ 0.108) traps the optimizer, propagating poor solutions through $\varepsilon_n = 2\text{--}13$. Cold start independently finds better basins at each point. Warm-start wins only 4/20 emittance points.

4 Parameter Sensitivity

4.1 1D Scans (S4)

Three 1D sweeps at the 0.5 ps operating point (500 particles, seed 42):

- **Energy spread** ($\sigma_E = 0.1\text{--}5\%$): all Excellent (MSE $< 10^{-3}$) up to $\sigma_E \approx 3\%$.
- **Chirp** ($h = 0\text{--}40 \times 10^9 \text{ s}^{-1}$): chirp has negligible effect on transverse matching (all Excellent).
- **Emittance** ($\varepsilon_n = 1\text{--}20$): the dominant sensitivity parameter. Fails at $\varepsilon_n \leq 2$ (beam too small) and $\varepsilon_n \geq 17$ (requires raised bounds).

4.2 2D Scans (S5)

Three 10×10 grids (100 points each, 500 particles):

- **S5a** ($\sigma_E \times h$) at $\varepsilon_n = 8$: 0 failures, 92 Excellent. Energy spread and chirp do not compromise matching.
- **S5b** ($\sigma_E \times \varepsilon_n$) at $h = 5 \times 10^9$: 18 failures at extreme emittances.
- **S5c** ($h \times \varepsilon_n$) at $\sigma_E = 0.5\%$: 20 failures at extreme emittances. Chirp does not rescue high-emittance failure.

The feasibility boundary is dominated by emittance.

4.3 Bunch Length Independence (S6, S9)

The 2 ps and 0.5 ps operating points produce identical quad currents. Root cause: transverse Twiss depends on ε and σ_δ , not σ_z ; bunch length enters only column 5 (time of flight), which `cal_twiss()` ignores. A 15-point bunch-length sweep (0.1 ps to 2.0 ps) confirms no degradation at any bunch length.

4.4 Verification (S7)

Re-running 7 emittance points and 3 energy-spread points at $N = 500, 1000, 2000$ particles reveals:

- Energy spread results are fully consistent across particle counts.
- Emittance results at extremes ($\varepsilon_n \neq 8$) are *not* robust — different particle samples lead to different NM basins in stages 1–10. Publication-quality claims at extreme emittances require multi-start + multi-seed statistics.

5 Longitudinal Dynamics

5.1 Transfer Map Analysis (W9)

COSY INFINITY 6D maps yield: $R_{56} = 27.09$ mm (elongates unchirped beams), $T_{566} = 0$ (no second-order path-length dependence). The transport line is intrinsically non-compressing.

5.2 Bunch Compression Feasibility (W12)

Can the transport line compress 2 ps $\rightarrow$ 0.5 ps?

- **Analytical compression**: floor ≈ 0.45 ps from $R_{56} \times \sigma_\delta$ at $\sigma_E = 0.5\%$.
- **Chirp**: $C = 4$ chirp gives ~ 0.67 ps, still above target.
- **Extended bounds** (15 A, 5 restarts): improves transverse MSE; σ_z unchanged.
- **Conclusion**: the transport line is not a compressor. Compression must occur upstream (velocity bunching or dedicated compressor).

5.3 σ_z Blowup Investigation (I6)

The 60–100 $\times$ longitudinal blowup originally reported was *not* reproduced with properly matched optics (MSE $\approx 2.3 \times 10^{-7}$). With optimised currents: ORDER 1 gives $\sigma_z = 2.264$ ps (1.13 $\times$), ORDER 3 gives $\sigma_z = 4.158$ ps (2.08 $\times$). The higher-order growth is correct physics from nonlinear path-length terms.

6 Chromatic Optimisation

Linear-optics matching ignores the momentum dependence of focusing strengths. At $\sigma_p = 0.5\%$, linear FELsim shows β_y errors up to 1466% compared with RF-Track’s intrinsic chromatic tracking (C8). After enabling chromatic mode for all elements (quads, dipole bodies, dipole wedges), a three-pass coordinate-descent re-optimisation with aperture losses yields:

- $\text{MSE} = 4.0 \times 10^{-4}$ ($\beta_x = 1.40$, $\alpha_x = 0.44$, $\beta_y = 0.23$, $\alpha_y = 0.03$)
- Transmission = 98.1% (up from 64.5% with linear-optimised currents)
- Significant current changes in the chicane region (elements 33–43) and downstream quads (70, 87, 93–97)

7 Beam Losses and Throughput

7.1 Loss Distribution (P1)

FELsim aperture tracking at the baseline operating point (2000 particles, $\sigma_p = 0.5\%$):

- Overall transmission: 64.5% (with linear-optimised currents)
- Dominant loss: FC1.DPW.111 (element 100, ± 6.3 mm) — 395 particles (56% of losses)
- Chicane exit dipoles (72–73, 89–90): 281 particles
- Quad losses minor: 34 particles total

7.2 Throughput Optimisation (W11)

Extending the Stage 11 objective to include transmission and bunch length via weighted scalar cost: $w_t \cdot \text{MSE}_{\text{Twiss}} + w_T \cdot (1 - T)^2 + w_\sigma \cdot (\sigma_t / \sigma_{\text{target}} - 1)^2$. Results (RF-Track with physical apertures):

- 2 ps transport: $\text{MSE} = 0.175$, $T = 42.8\%$, $\sigma_t = 1.83$ ps, $I_{\text{peak}} = 5.6$ A
- 0.5 ps compression: $\text{MSE} = 0.071$, $T = 28.4\%$, $\sigma_t = 1.31$ ps (target 0.5), $I_{\text{peak}} = 5.2$ A

Compression failed — σ_t target unreachable with Stage 11 quads alone, consistent with the W12 feasibility finding.

8 FR3+MGE Optimisation (C3)

Matching the beamline with COSY INFINITY’s 3rd-order fringe fields plus measured fieldmaps (MGE) requires simultaneous optimisation of all 23 quadrupole currents. Sequential FIT fails because the per-stage local optima under FR3+MGE are incompatible with the global solution.

Four CMA-ES campaigns on the Koa HPC cluster:

- **v1**: 10,020 evals, $\text{MSE} = 1030$ (barely unstable, $\cos \mu \approx 1.03$).
- **v2**: 16,000 evals, sigma collapsed to 1.3×10^{-5} ; no stable solution found.
- **v3**: Killed after 247 evals.
- **v4** (current): Capped objective ensuring any stable solution beats any unstable one. 663 evals completed, best objective 12,009 (unstable but steadily improving).

Root cause: the MSE landscape has an extremely narrow stability basin in 23D. The capped objective (v4) directly addresses the pathology where unstable solutions could outcompete stable ones with poor Twiss.

9 Infrastructure

Element labels (I1). All beamline elements carry human-readable names propagated from Excel (Label/Nomenclature columns) and JSON/YAML (`name` field) through the entire pipeline.

PALS alignment (I3, I8). The v3 lattice format adds PALS-aligned fields (`Bn1`, `BendP`) with full backward compatibility. The standalone `pals2cosy` converter (v0.3.0) supports both official PALS and FELsim v2/v3 formats.

Test suite (C4, I6). 210 tests (7 skipped), covering: unit tests for transfer matrices and Twiss computation; frozen regression tests for the full beamline; cross-code benchmarks (FELsim vs RF-Track); adapter round-trips (Excel/JSON/YAML); chromatic physics (30 tests); attribute guards (AST-based typo detection).

Multi-code framework (I7). The `MultiCodeSimulator` enables hybrid simulations with per-section code selection and configuration passthrough (space charge, apertures). Validated for FELsim+RF-Track, FELsim+COSY, COSY+RF-Track, and three-way hybrids.

10 Summary of Findings

1. The 11-stage sequential NM optimisation reliably matches the undulator Twiss targets at the baseline operating point, with cross-code agreement to $\text{MSE} \sim 10^{-6}$ – 10^{-9} .
2. Two-phase NM→CMA-ES polishing improves Stage 11 precision by up to $10^{10}\times$ and is integrated as the default post-processing step.
3. The feasibility boundary is dominated by normalised emittance (ε_n). Energy spread (σ_E) and chirp (h) have negligible impact at the baseline $\varepsilon_n = 8 \pi \cdot \text{mm} \cdot \text{mrad}$.
4. Bunch length does not affect transverse matching (confirmed analytically and numerically).
5. Bunch compression below ~ 0.45 ps is not achievable by the transport line alone ($R_{56} = +27$ mm, $T_{566} = 0$).
6. Chromatic optimisation with aperture tracking raises transmission from 64.5% to 98.1% with $\text{MSE} = 4.0 \times 10^{-4}$.
7. The Stage 11 landscape is unimodal; robustness issues at extreme emittances originate from upstream stages, not the final matching section.
8. Sequential warm-starting is counterproductive for this beamline (basin trapping).
9. FR3+MGE optimisation requires simultaneous 23-variable search (in progress on HPC).

11 Study Index

ID	Title	Status	Key Result
S1	2 ps Baseline Optimisation	Done	$\text{MSE} < 10^{-5}$
S2	0.5 ps Emittance Conservation	Done	Joint 4-var Stage 11
S3	0.5 ps Paper-Aligned	Done	Identical to S1 currents
S4	1D Parameter Sensitivity	Done	Emittance dominates
S5	2D Coupled Scans	Done	σ_E , h negligible
S6	Bunch Length Sensitivity	Done	No effect on matching
S7	Verification Runs	Done	Extremes not robust

ID	Title	Status	Key Result
S8	Multi-Start Robustness	Done	Stage 11 unimodal
S9	Bunch Length Independence	Done	Linear decoupling
W1	Chirp Comparison	Done	h negligible on Twiss
W2	Emittance Multi-Start	Done	Dips at $\varepsilon_n = 14\text{--}16$ resolved
W4	COSY Cross-Validation	Done	FR1 MSE = 3.7×10^{-9}
W5	Chirp Assessment	Done	$h = 5 \times 10^9$ is low-moderate
W6	Glyfada Benchmark	Done	NM wins by $10^3\text{--}10^6 \times$
W7	CMA-ES Re-Benchmark	Done	CMA-ES wins by $10^9\text{--}10^{10} \times$
W9	COSY Longitudinal	Done	$R_{56} = +27$ mm, $T_{566} = 0$
W10	Beam Losses Study	Done	40–50% loss at 0.5 ps
W11	Throughput Optimisation	Done	$T = 42.8\%$ at 2 ps
W12	Compression Feasibility	Done	Floor ≈ 0.45 ps
C1	RF-Track Cross-Validation	Done	5-way comparison
C3	FR3+MGE Optimisation	In Progress	CMA-ES v4 on Koa
C5	RF-Track SBend Bug	Done	Analytical workaround
C8	Chromatic Quad Matrices	Done	0.00% vs RF-Track
O1	Warm-Starting	Done	Counterproductive
O3	Glyfada Adapter	Done	<code>method='glyfada'</code>
O5	CMA-ES Polishing	Done	Integrated in production
P1	Aperture Loss Tracking	Done	64.5% baseline T
P2	Chromatic DPW Edges	Done	Symplectic, vectorised
P3	Chromatic Dipole Body	Done	$< 1\%$ at $\sigma_p = 0.5\%$
I1	Element Labels	Done	End-to-end propagation
I3	PALS2COSY Converter	Done	v0.3.0 (55 tests)
I7	Multi-Code Framework	Done	3-way hybrid validated
I8	v3 Lattice Format	Done	PALS-aligned fields

References

- [1] A. Weinberg, A. Fisher, and R. Li, “Design of a compact FEL oscillator for the University of Hawai‘i,” arXiv:2510.14061v1, 2025.